RELATIVE PSEUDO-COHERENCE IN LOW DEGREES FOR MODULES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 17052-17067, immediately following the statement at 17039-17050.

Editorial note. Three cross-chapter pointers (Cohomology, Modules, Morphisms) are recorded in assembly.external_references.

Editorial note. The author's closing sentence 'The proof of (2) is similar but less involved.' is retained verbatim.

Allowed standard prerequisites (editorial).

- Pseudo-coherence of a module over a polynomial algebra. Cohomology, lemma cohomology-lemma-finite-cohomology, cited at proof line 17060. Identifies (-1)-pseudo-coherence of the pushed-forward module with finite presentation over A^n_S.
- Finitely presented modules are quasi-coherent. Modules, lemma modules-lemma-finite-presentation-quasi-coherent, cited at proof lines 17061-17062. Supplies one direction of the quasi-coherence clause.
- Quasi-coherence is preserved and reflected by a closed immersion. Morphisms, lemma morphisms-lemma-i-star-equivalence, cited at proof lines 17063-17064. Transfers quasi-coherence between F and i_*F, which closes part (3).
- Definition of m-pseudo-coherence relative to the base. used at proof line 17053 for part (1). Part (1) is immediate from the definition, as the author states.

Source reference. Original reference 08DN: focused-seeds/geometry/S9-morph-rel-n-pseudo-module/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01BO: focused-seeds/geometry/S9-morph-rel-n-pseudo-module/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01QY: focused-seeds/geometry/S9-morph-rel-n-pseudo-module/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $X \to S$ be a morphism of schemes which is locally of finite type. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module. Then*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

(1) *$\mathcal{F}$ is m-pseudo-coherent relative to S for all $m > 0$,*
(2) *$\mathcal{F}$ is 0-pseudo-coherent relative to S if and only if $\mathcal{F}$ is a finite type $\mathcal{O}_X$-module,*
(3) *$\mathcal{F}$ is (-1)-pseudo-coherent relative to S if and only if $\mathcal{F}$ is quasi-coherent and finitely presented relative to S.*

Proof. Part (1) is immediate from the definition. To see part (3) we may work locally on X (both properties are local). Thus we may assume X and S are affine. Choose a closed immersion $i : X \to \mathbf{A}^n_S$. Then we see that $\mathcal{F}$ is (-1)-pseudo-coherent relative to S if and only if $i_*\mathcal{F}$ is (-1)-pseudo-coherent, which is true if and only if $i_*\mathcal{F}$ is an $\mathcal{O}_{\mathbf{A}^n_S}$-module of finite presentation, see Cohomology, Lemma 08DN. A module of finite presentation is quasi-coherent, see Modules, Lemma 01BO. By Morphisms, Lemma 01QY we see that $\mathcal{F}$ is quasi-coherent if and only if $i_*\mathcal{F}$ is quasi-coherent. Having said this part (3) follows. The proof of (2) is similar but less involved. □